浙江大学 ZHEJIANG UNIVERSITY

离散数学及其应用 · 期末复习 · Discrete Mathematics

期末复习讲义
与真题巩固

知识精要 ＋ 分章节历年真题 · 读练一体

使用方式：

本册覆盖第 1–11 章（不含第 7 章）。每章先读「**知识精要**」梳理考点、定义与公式，随后立即进入「**真题巩固**」完成对应章节的历年期末真题，做到**看完一章、练完一章**。

知识点以中文讲解、**关键术语保留英文**；真题均按原卷的全英文形式给出（仅作排版与符号规范化），**不附解答**，留作自测。题目选自浙大《离散数学及其应用》历年期末卷（2003、2019–2025）。

内容已对照**《2025–2026 春夏学习内容及考试范围》**（教材 Rosen 第 8 版），剔除大纲中标 * 的不考小节（如 4.2、4.5/4.6、5.5、6.6、8.3、9.2、2.6 等）。

目　录

1 逻辑 Logic 1
2 集合、函数与数列 Sets, Functions & Sequences 4
3 算法 Algorithms 7
4 数论 Number Theory 8
5 归纳与递归 Induction & Recursion 10
6 计数 Counting 11
8 高级计数 Advanced Counting 14
9 关系 Relations 17
10 图 Graphs 20
11 树 Trees 23

1 逻辑 Logic

第 1 章 逻辑

Logic：命题逻辑 · 谓词与量词 · 范式 · 推理

1

知识精要 逻辑是全卷的“语言”：先掌握联结词与等价式，再处理量词，最后是范式与推理。

本章主线

1. **命题逻辑 (Propositional Logic)**：联结词 $\neg, \wedge, \vee, \oplus, \to, \leftrightarrow$、真值表、逻辑等价。
2. **谓词与量词 (Predicates & Quantifiers)**：$\forall, \exists$、嵌套量词、量词的 De Morgan 律。
3. **范式 (Normal Forms)**：主析取范式 (full DNF)、主合取范式 (full CNF)、前束范式。
4. **推理 (Inference)**：推理规则、有效论证、谓词逻辑推理。

1.1 联结词与蕴含

联结词与等价（必记） **蕴含 (implication)**：$p \to q \equiv \neg p \vee q$；其否定 $\neg(p \to q) \equiv p \wedge \neg q$。
逆否 (contrapositive) $\neg q \to \neg p \equiv p \to q$；**逆 (converse)** $q \to p$；**否 (inverse)** $\neg p \to \neg q$（逆与否互为逆否，彼此等价）。
双条件：$p \leftrightarrow q \equiv (p \to q) \wedge (q \to p)$；**异或**：$p \oplus q \equiv (p \wedge \neg q) \vee (\neg p \wedge q) \equiv \neg(p \leftrightarrow q)$。

重要逻辑等价式

- De Morgan：$\neg(p \wedge q) \equiv \neg p \vee \neg q$，$\neg(p \vee q) \equiv \neg p \wedge \neg q$。
- 分配律：$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$；$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$。
- 吸收律：$p \vee (p \wedge q) \equiv p$；$p \wedge (p \vee q) \equiv p$。
- 重言/矛盾：$p \vee \neg p \equiv \mathbf{T}$，$p \wedge \neg p \equiv \mathbf{F}$。

可满足性 (satisfiability)：存在一组真值赋值使公式为真；重言式 (tautology) 恒真，矛盾式 (contradiction) 恒假。

1.2 谓词、量词与嵌套

量词的等价与否定 **量词 De Morgan**：$\neg\forall x P(x) \equiv \exists x\, \neg P(x)$，$\neg\exists x P(x) \equiv \forall x\, \neg P(x)$。
与无关命题：若 Q 不含 x，则 $\forall x(P(x) \vee Q) \equiv \forall x P(x) \vee Q$，$\forall x P(x) \to Q \equiv \exists x(P(x) \to Q)$。
分配性：$\forall x(P \wedge Q) \equiv \forall x P \wedge \forall x Q$（$\forall$ 配 $\wedge$）；$\exists x(P \vee Q) \equiv \exists x P \vee \exists x Q$（$\exists$ 配 $\vee$）。其余方向一般只有单向蕴含。

量词顺序不可随意交换 $\forall x \exists y\, P(x,y)$ 与 $\exists y \forall x\, P(x,y)$ **不等价**：后者更强，蕴含前者，反之不然。嵌套量词的否定要**逐层取反并交换量词**：$\neg\forall x \exists y P \equiv \exists x \forall y\, \neg P$。

1.3 范式：主析取与主合取

主析取范式与主合取范式 **极小项 (minterm)**：每个变元恰出现一次（原变量或其否定）的合取，对应使其为真的**唯**一赋值。
极大项 (maxterm)：每个变元恰出现一次的析取，对应使其为假的唯一赋值。
主析取范式 (full DNF) = 所有使公式**为真**的极小项之析取，记 $\Sigma(\dots)$；
主合取范式 (full CNF) = 所有使公式**为假**的极大项之合取，记 $\Pi(\dots)$。
二者下标互补：使公式真的行进 Σ，使公式假的行进 Π。

由真值表写范式 / 化前束范式 列真值表 ⇒ 真行收集极小项得主析取范式，假行收集极大项得主合取范式。
谓词公式化**前束范式 (prenex normal form)**：先把所有量词提到最前（必要时改名以避免变量捕获），再处理母式。

1.4 推理规则

常用推理规则 (Rules of Inference) 假言推理 (modus ponens)：$p,\ p \to q \vdash q$；　拒取式 (modus tollens)：$\neg q,\ p \to q \vdash \neg p$。
假言三段论：$p \to q,\ q \to r \vdash p \to r$；　析取三段论：$p \vee q,\ \neg p \vdash q$。
消解 (resolution)：$p \vee q,\ \neg p \vee r \vdash q \vee r$。
谓词：全称/存在的**实例化 (instantiation)** 与**一般化 (generalization)**。

证明方法 (Proof Methods) **直接证明**：由前提逐步推出结论。**逆否 (contraposition)**：证 $\neg q \to \neg p$ 代替 $p \to q$。**反证 (contradiction)**：设结论不成立导出矛盾。
分情形 (proof by cases)；**举反例 (counterexample)** 否定全称命题；存在性分**构造性/非构造性**证明，唯一性 = 存在 + 唯一。

✎ **真题巩固** 命题逻辑 · 等价式、量词、范式与推理

真题 1.1 2025 Find the full conjunctive normal form of $(p \oplus q) \to r$.

真题 1.2 2025 (Multiple choice) Which of the following statements are true? (Assume the same domain for all variables.) (A) $\forall x(P(x) \wedge Q(x)) \equiv \forall x P(x) \wedge \forall x Q(x)$; (B) $\exists x(P(x) \wedge Q(x)) \equiv \exists x P(x) \wedge \exists x Q(x)$; (C) $\neg\exists x(P(x) \wedge Q(x)) \equiv \forall x(\neg P(x) \vee \neg Q(x))$; (D) $\exists x(P(x) \to Q(x)) \equiv \exists x P(x) \to \exists x Q(x)$.

真题 1.3 2019 (True/False) If $\forall x(P(x) \vee Q(x))$ and $\forall x\big((\neg P(x) \wedge Q(x)) \to R(x)\big)$ are true, then $\forall x(\neg R(x) \to P(x))$ is also true (domains of all quantifiers are the same).

真题 1.4 2019 (Fill in the blank) The full disjunctive normal form of $\neg r \vee (p \leftrightarrow q)$ is _____.

真题 1.5 2019 秋冬 (True/False) Let φ be a proposition with three variables p, q, r. If exactly one of the assignments $000, 011, 100, 111$ makes φ true, then φ can be written in full disjunctive normal form $\Sigma(1, 2, 5, 6)$.

真题 1.6 2019 秋冬 (True/False) $\forall x P(x) \to \exists y Q(y) \ \equiv\ \forall x \exists y\big(P(x) \to Q(y)\big)$.

真题 1.7 2019 秋冬 Construct a logical argument using rules of inference to show that the following premises imply the conclusion: *It rained.* Premises: If it does not rain or if it is not

foggy, then the sailing race will be held and the life-saving demonstration will go on. If the sailing race is held, then the trophy will be awarded. The trophy was not awarded. Justify each step by indicating the rule applied.

真题 1.8 2021 (True/False) $\forall x P(x) \to Q \equiv \forall x\big(P(x) \to Q\big)$, where Q is independent of x.

真题 1.9 2021 (True/False) If the assignments $000, 011, 110$ make the propositional formula φ false, then φ can be converted into full conjunctive normal form $\Pi(0, 3, 6)$.

真题 1.10 2021 Construct an argument to prove that the following reasoning is valid. Hypotheses: $\neg p \vee q \to r$, $s \vee \neg q$, $\neg t$, $p \to t$, $\neg p \wedge r \to \neg s$. Conclusion: $\neg q$.

真题 1.11 2020 春夏 (Fill in the blank) The full disjunctive normal form of $\neg r \vee (p \leftrightarrow q)$ is _____.

真题 1.12 2020 春夏 Suppose $Q(a, b, c, x)$ is a quadratic equation of the form "$ax^2+bx+c=0$" with $a \neq 0$. Use quantifiers and logical connectives to express the fact that any $Q(a, b, c, x)$ has at most two real roots.

真题 1.13 2003 (True/False) $\forall x P(x) \wedge \exists x Q(x)$ and $\forall x \exists y\big(P(x) \wedge Q(y)\big)$ are logically equivalent.

真题 1.14 2003 Let the proposition formula $G = p \wedge (q \leftrightarrow r)$. (a) Find the full disjunctive normal form of G. (b) Display all the assignments that make G false.

2 集合、函数与数列 Sets, Functions & Sequences

第 2 章　集合、函数与数列

Sets, Functions & Sequences：集合运算 · 映射 · 基数与可数性　2

知识精要　把集合恒等式、函数三性质、可数性串成一条线；可数性证明是高频压轴。

本章主线

1. **集合与运算**：子集、幂集、笛卡尔积、$\cup\cap-\triangle$、集合恒等式。
2. **函数**：单射 (injection)、满射 (surjection)、双射 (bijection)、复合与反函数、取整函数。
3. **序列与求和**：等差/等比、求和公式。
4. **基数 (cardinality)**：可数 (countable) 与不可数、对角线法、相等基数的构造。

2.1 集合运算与恒等式

幂集 · 笛卡尔积 · 恒等式　$|P(A)|=2^{|A|}$；$|A\times B|=|A|\cdot|B|$。
De Morgan：$\overline{A\cup B}=\bar{A}\cap\bar{B}$，$\overline{A\cap B}=\bar{A}\cup\bar{B}$；分配律、吸收律与逻辑等价式同构。
对称差 (symmetric difference)：$A\triangle B=(A-B)\cup(B-A)$，满足结合律，且 $A\triangle C=B\triangle C\Rightarrow A=B$（消去律）。

2.2 函数的三种性质

单射 · 满射 · 双射　$f:A\to B$。**单射**：$f(x_1)=f(x_2)\Rightarrow x_1=x_2$；**满射**：$\forall y\in B\,\exists x,\ f(x)=y$；**双射**：既单又满，存在反函数 f^{-1}。
从 A（$|A|=m$）到 B（$|B|=n$）：函数共 n^m 个；单射 $n(n-1)\cdots(n-m+1)$ 个（$m\le n$）。
像的性质：$f(S\cap T)\subseteq f(S)\cap f(T)$（一般**不取等**）；$f(S\cup T)=f(S)\cup f(T)$。

取整函数 Floor & Ceiling　$\lfloor x\rfloor$ 为不超过 x 的最大整数，$\lceil x\rceil$ 为不小于 x 的最小整数。
常用：$\lceil n/k\rceil=\lfloor(n-1)/k\rfloor+1$（$n,k$ 正整数）；$\lfloor x\rfloor+\lfloor x+\frac{1}{2}\rfloor=\lfloor 2x\rfloor$。

2.3 序列与求和

常用求和　等比：$\displaystyle\sum_{k=0}^{n}ar^k=\frac{a(r^{n+1}-1)}{r-1}\ (r\neq1)$，$\displaystyle\sum_{k=0}^{\infty}ar^k=\frac{a}{1-r}\ (|r|<1)$。

$$\sum_{k=1}^{n}k=\frac{n(n+1)}{2},\quad\sum_{k=1}^{n}k^2=\frac{n(n+1)(2n+1)}{6},\quad\sum_{k=1}^{n}k^3=\left(\frac{n(n+1)}{2}\right)^2.$$

2.4 基数与可数性

可数集 (Countable Sets)　$|A|=|B|\iff$ 存在双射 $A\to B$。可数 = 有限或可数无穷（与 $\mathbb{Z}^+$ 等势）。
$\mathbb{Z},\mathbb{Q}$，$\mathbb{Z}\times\mathbb{Z}$ 均可数无穷；$\mathbb{R}$、$(0,1)$、$P(\mathbb{N})$、$\mathbb{N}\to\{0,1\}$ 的全体函数**不可数**（Cantor 对角线法）。
Schröder–Bernstein：若存在单射 $A\to B$ 与 $B\to A$，则 $|A|=|B|$。证“同基数”常**显式构造双射**（如线性映射 $(0,1)\to(a,b)$）。

幂集与“属于/包含” $P(A)=P(B) \iff A=B$；$P(A)\subseteq P(B) \iff A\subseteq B$。但 $P(A)\in P(B)$ **不能**推出 $A\in B$。区分 $\subseteq$（子集）与 $\in$（元素）：可以同时有 $A\subseteq B$ 且 $A\in B$（如 $A=\{\emptyset\}, B=\{\emptyset,\{\emptyset\}\}$）。

✎ 真题巩固　集合 · 函数 · 基数与可数性

真题 2.1 2025 (True/False) The function $f:\mathbb{Z}\times\mathbb{Z}\to\mathbb{Z}$ where $f(n,m)=5n+3m$ is onto and one-to-one.

真题 2.2 2025 (True/False) If A,B,C are sets, then $(A-B)\cup(A-C)=A \iff A\cap B\cap C=\emptyset$.

真题 2.3 2025 (Multiple choice) Suppose $f:\mathbb{R}\to\mathbb{R}$ is a function and for every $x,y\in\mathbb{R}$, $f(x+y)=f(x)+f(y)$. Which statements are true? (A) $f(0)=0$; (B) $f(1)=0$; (C) for every $n\in\mathbb{N}$, $f(n)=n\cdot f(1)$; (D) None of above.

真题 2.4 2021 (True/False) Let A,B,C be arbitrary sets. If $A-C\subseteq B-C$, then $A\cup C\subseteq B\cup C$.

真题 2.5 2021 (True/False) Let A,B be two sets. If $\rho(A)\subseteq\rho(B)$, then $A\subseteq B$, where $\rho(X)$ is the power set of X.

真题 2.6 2021 (True/False) The set of all functions from $\mathbb{N}$ to $\{0,1\}$ is countably infinite.

真题 2.7 2021 Let A be an arbitrary infinite set, B a countably infinite set, and $A\cap B=\emptyset$. Prove that the sets A and $A\cup B$ have the same cardinality.

真题 2.8 2019 (True/False) $P(A)=P(B)$ if and only if $A=B$, where $P(X)$ is the power set of X.

真题 2.9 2019 (True/False) There is no countably infinite set A with a bijection $A\to A\times A$.

真题 2.10 2019 秋冬 (True/False) There are two sets A and B such that both $A\subseteq B$ and $A\in B$.

真题 2.11 2019 秋冬 (True/False) Let A,B be two sets. If $2^A\subseteq 2^B$, then $A\subseteq B$, where 2^X is the power set of X.

真题 2.12 2019 秋冬 Prove that the two sets of real numbers $(0,1)$ and $[a,b) = \{x \in \mathbb{R} \mid a \leq x < b\}$ have the same cardinality.

真题 2.13 2020 春夏 (True/False) Let f be a function from set A to set B, and S, T subsets of A. Then $f(S \cap T) \subseteq f(S) \cap f(T)$.

真题 2.14 2020 春夏 (True/False) Let A, B, C be sets and $\triangle$ the symmetric difference. If $A \triangle C = B \triangle C$, then $A = B$.

真题 2.15 2020 春夏 Suppose $Q(a,b,c,x)$ is a quadratic equation "$ax^2 + bx + c = 0$", where $a(\neq 0), b, c$ are integers and x is real. Determine whether the set of real roots of all such $Q(a,b,c,x)$ is countable, and explain why.

真题 2.16 2003 (True/False) $f(n) = \lceil n/2 \rceil$ is a one-to-one function (injection) from $\mathbb{Z}$ to $\mathbb{Z}$.

真题 2.17 2003 (True/False) Let A, B be any sets. If $P(A) \in P(B)$, then $A \in B$, where $P(S)$ is the power set of S.

真题 2.18 2003 (Fill in the blank) Let $A = \{2,3,4\}, B = \{1,2\}, C = \{4,5,6\}$. The result of $(A \oplus B) \oplus (B \oplus C)$ is _____, where $\oplus$ is the symmetric difference.

3 算法 Algorithms

第 3 章 算法

Algorithms：搜索与排序 · 贪心 · 函数增长 Big-O

3

知识精要 本章在期末多以“复杂度判断”形式融入其它题；核心是 Big-O 与常见复杂度阶。

本章主线

1. **典型算法**：线性/二分搜索、冒泡/插入排序、贪心 (greedy)。
2. **函数增长**：Big-O、Big-Ω、Big-Θ 的定义与运算。
3. **复杂度**：最好/平均/最坏情形，基本语句计数规则。

3.1 典型算法的复杂度

搜索与排序 线性搜索 (linear search) $O(n)$；二分搜索 (binary search) $O(\log n)$（要求有序）。冒泡排序 (bubble sort)、插入排序 (insertion sort)：最坏与平均均 $O(n^2)$；归并/快排平均 $O(n\log n)$。
贪心算法：每步取局部最优；正确性需证明（如美元硬币系统贪心最优，但任意面值系统未必）。

3.2 Big-O 与渐近阶

Big-O / Big-Ω / Big-Θ $f=O(g)$：$\exists C,k,\ \forall x>k,\ |f(x)|\le C|g(x)|$（上界）。
$f=\Omega(g)$：$\exists C,k,\ |f(x)|\ge C|g(x)|$（下界）；$f=\Theta(g)$：既 O 又 Ω（同阶）。
常见阶（升序）：$1<\log n<n<n\log n<n^2<2^n<n!$。多项式 $a_dn^d+\cdots$ 为 $\Theta(n^d)$。

最坏情形与算法选择 复杂度通常指**最坏情形**的增长阶；同一算法在不同输入下差异大。Dijkstra 用二叉堆为 $O((n+m)\log n)$，朴素数组实现为 $O(n^2)$——叙述复杂度时要说明实现与计量方式。

✎ **真题巩固** 算法 · 复杂度判断（本章在历年卷中多融入图、递归等综合题）

真题 3.1 2025 (True/False) The worst-case time complexity of Dijkstra's algorithm for computing the shortest path between two points in a weighted graph with n vertices and m edges is $O(nm)$.

真题 3.2 2023 (True/False) If $f(1)=1$ and $f(n)=2f(\lfloor n/2\rfloor)$, then $f(n)\le n$.

4 数论 Number Theory

第 4 章 数论

Number Theory：整除与同余 · 素数与 GCD · 同余方程 4

知识精要 核心是“同余 + 欧几里得”：模运算、逆元、CRT 与费马小定理构成主干。

本章主线

1. **整除与同余**：$a \mid b$、带余除法、模运算与同余的运算性质。
2. **素数与 GCD/LCM**：算术基本定理、欧几里得算法、Bézout 定理。
3. **同余方程**：模逆元 (inverse modulo m)、线性同余、中国剩余定理 (CRT)、费马小定理。

4.1 整除、同余与素因子分解

同余的运算性质 $a \equiv b \pmod m \iff m \mid (a-b)$。若 $a \equiv b,\ c \equiv d \pmod m$，则

$$a + c \equiv b + d, \qquad ac \equiv bd \pmod m.$$

求 $a^n \bmod m$ 时，常借助下方**费马小定理**把指数降下来再算（如 $7^{222} \bmod 11$）。

算术基本定理与约数个数 每个正整数唯一分解为 $n = p_1^{e_1} p_2^{e_2} \cdots p_k^{e_k}$。
正约数个数 $= (e_1+1)(e_2+1)\cdots(e_k+1)$；gcd 取各素数幂的较小指数、lcm 取较大指数，且 $\gcd(a,b) \cdot \operatorname{lcm}(a,b) = ab$。

4.2 欧几里得算法与逆元

Euclid · Bézout · 逆元 $\gcd(a,b) = \gcd(b, a \bmod b)$（辗转相除）。**Bézout**：$\gcd(a,b) = sa+tb$，由扩展欧几里得求 s,t。
模逆元：a^{-1} 存在 $\iff \gcd(a,m)=1$，由 $sa+tm=1$ 得 $a^{-1} \equiv s \pmod m$。
解 $ax \equiv b \pmod m$：两边乘 a^{-1} 得 $x \equiv a^{-1}b$。

CRT 与费马小定理 **中国剩余定理**：$m_1, \ldots, m_k$ 两两互素时，方程组 $x \equiv a_i \pmod{m_i}$ 在模 $M = \prod m_i$ 下有唯一解。
费马小定理：p 素数且 $p \nmid a$，则 $a^{p-1} \equiv 1 \pmod p$，故 $a^n \bmod p$ 可把指数对 $p-1$ 取模化简。

✎ **真题巩固** 数论 · 约数、同余、逆元与中国剩余定理

真题 4.1 2025 (True/False) The quadratic equation $ax^2+bx+c=0$ has no rational solutions when a, b, c are odd integers.

真题 4.2 2025 (Fill in the blank) How many positive divisors does 2025 have?

真题 4.3 2025 (Fill in the blank) There are three consecutive positive integers that are divisible by 7, 5 and 11 respectively. Find the smallest positive integer solution.

真题 4.4 **2023** $7^{222} \equiv x \pmod{11}$. Find x.

真题 4.5 **2023** Solve $29x \equiv 1 \pmod{37}$; find the smallest positive integer x satisfying the equation.

真题 4.6 **2023** (True/False) For positive integers n and k, $\lceil n/k \rceil = \lfloor (n-1)/k \rfloor + 1$.

5 归纳与递归 Induction & Recursion

第 5 章 归纳与递归

Induction & Recursion：数学归纳法 · 结构归纳 · 递归算法 5

知识精要 把“归纳假设用在哪一步”说清楚是得分关键；强归纳与良序性是同一思想的不同外衣。

本章主线

1. **数学归纳法**：基础步 + 归纳步；**强归纳法 (strong induction)** 与**良序性 (well-ordering)**。
2. **递归定义**：递归函数、递归集合、合式公式、树。
3. **结构归纳法 (structural induction)**：对递归构造的对象作归纳。
4. **递归算法**：递归与迭代、归并排序、正确性。

5.1 归纳法的三种形态

数学归纳 / 强归纳 / 良序 **弱归纳**：证 $P(n_0)$，且 $P(k)\Rightarrow P(k+1)$。
强归纳：证 $P(n_0)$，且 $\big[P(n_0)\wedge\cdots\wedge P(k)\big]\Rightarrow P(k+1)$——适合“依赖多个前项”的递推。
良序性：$\mathbb{N}$ 的每个非空子集有最小元，常用于“取最小反例”反证。

递归定义与结构归纳 递归定义含**基础条款**（基本元素）与**递归条款**（由旧造新）。
结构归纳：在基础元素上验证 P，并证明每条递归规则保持 P。常用于字符串、合式公式、树的性质（如满二叉树顶点数 $\le 2^{h+1}-1$）。

归纳常见失分点 基础步不可省；强归纳要确保**基础步覆盖足够多的起始情形**（如 a_n 依赖 a_{n-1},a_{n-2} 需验证两个起点）；归纳步必须**显式引用归纳假设**。

✎ **真题巩固** 归纳与递归 · 不等式归纳、递推关系判别

真题 5.1 2019 (True/False) Let $a_1=2$, $a_2=9$, and $a_n=2a_{n-1}+3a_{n-2}$ for $n\ge 3$. Then $a_n\le 3^n$ for all positive integers n.

真题 5.2 2023 (True/False) $D_n=(n-1)(D_{n-1}-D_{n-2})$ is a linear homogeneous constant-coefficient recurrence relation.

真题 5.3 2003 (True/False) The recurrence relation $D_n=(n-1)(D_{n-1}+D_{n-2})$ is a linear homogeneous recurrence with constant coefficients.

真题 5.4 2025 Given Fibonacci numbers f_n with $f_1=f_2=1$, and a sequence a_n satisfying $a_n=3a_{n-1}-a_{n-2}$ $(n\ge 3)$ with $a_1=1, a_2=3$, use induction to show that $a_n=f_{2n}$. *(此为 2025 卷综合题的归纳部分，完整题见第 8 章真题。)*

6 计数 Counting

第 6 章 计数

Counting：基本原理 · 鸽巢 · 排列组合 · 隔板法 6

知识精要 先分清“有序/无序、可重/不可重、可区分/不可区分”，再选公式；鸽巢题重在“构造抽屉”。

本章主线

1. **基本原理**：加法法则、乘法法则、二集合容斥、除法法则。
2. **鸽巢原理 (Pigeonhole)**：基本与广义形式、典型构造。
3. **排列组合**：$P(n,r), C(n,r)$、二项式定理、Pascal 恒等式。
4. **广义排列组合**：可重排列、可重组合（隔板法）、球盒问题。

6.1 基本原理与鸽巢

加法 · 乘法 · 容斥 · 鸽巢 加法法则 (不相交并)、乘法法则 (步骤独立)。二集合容斥 $|A\cup B| = |A| + |B| - |A\cap B|$。
鸽巢原理：$n+1$ 个物体放入 n 个抽屉，必有抽屉含 ≥ 2 个。
广义鸽巢：N 个物体放入 k 个抽屉，必有抽屉含 $\geq \lceil N/k \rceil$ 个。

6.2 排列、组合与二项式

排列 · 组合 · 二项式定理

$$P(n,r) = \frac{n!}{(n-r)!}, \qquad C(n,r) = \binom{n}{r} = \frac{n!}{r!(n-r)!}.$$

二项式定理：$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^{n-k} y^k$。
Pascal 恒等式 $\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$；$\sum_k \binom{n}{k} = 2^n$。

广义排列组合 · 球盒问题 可重排列：从 n 类取 r 个排成序，n^r。
可重组合 (隔板法)：n 类中取 r 个的多重集数 $= \binom{n+r-1}{r}$；等价于 $x_1 + \cdots + x_n = r$ 的非负整数解数。
含不可区分对象的排列：$\dfrac{n!}{n_1!n_2!\cdots n_k!}$。
球盒小结：球可/不可区分 × 盒可/不可区分，配合“空盒/非空”约束分类讨论。

“至少/恰好/不在原位” “恰好”常用容斥；“不在原位”是错位排列 (derangement)，见第 8 章。带上下界的整数解先**换元去界**再隔板。鸽巢题先想清楚“物体是什么、抽屉怎么分”。

✎ **真题巩固** 计数 · 排列组合、整数解、鸽巢原理

真题 6.1 2025 (Fill in the blank) Find the coefficient of x^5 in the expansion of $\left(x - \dfrac{2}{x}\right)^9$.

真题 6.2 2025 (Fill in the blank) How many ways are there to distribute 6 distinct apples to John, Alice, Tom and Sunny, such that exactly one person receives no apples?

真题 6.3 2025 (Fill in the blank) If A, B, C are sets with $A \cup B \cup C = \{1, 2, 3, 4, 5, 6\}$, $A \cap B = \{5\}$, and $B \cap C = \{2\}$, find the number of all possible cases of the sets A, B, C.

真题 6.4 2025 (Multiple choice) Let $S = \{x \in \mathbb{N} \mid 1 \leq x \leq 2025\}$. At most how many numbers can be selected from S to form a subset $T \subseteq S$, such that for any two different numbers $x, y \in T$, their sum is not divisible by their absolute difference? (A) 1012; (B) 675; (C) 506; (D) 405.

真题 6.5 2019 How many different ways can you put 9 coins in 9 boxes labeled $B_1, \ldots, B_9$: (1) coins all different and no box is empty? (2) coins all different and only two boxes B_1 and B_9 are empty? (3) coins all different and exactly four boxes are not empty? (4) coins all different and each box is either empty or contains exactly three coins? (5) coins identical? (6) coins identical and exactly six boxes are empty?

真题 6.6 2019 (Fill in the blank) Suppose $S = \{a, b\}$. How many ordered pairs (A, B) are there such that A, B are subsets of S with $A \subseteq B$?

真题 6.7 2020 春夏 Selecting 3 numbers from $1, 2, 3, \ldots, 100$ such that the sum of these 3 numbers is divisible by 4. Find the total number of different selections.

真题 6.8 2020 春夏 Assume you have 20 balls and three boxes (labeled A, B, C). (a) Assuming the balls are distinguishable, in how many ways can you put the balls into the boxes? (b) Assuming the balls are identical and each box must have at least two balls, in how many ways?

真题 6.9 2003 How many solutions are there to $x_1 + x_2 + x_3 = 12$, where x_1, x_2, x_3 are nonnegative integers with: (a) $x_1 > 1$, $x_2 < 3$, $x_3 > 4$? (b) x_1 and x_2 being odd numbers, and $x_3 > 5$?

真题 6.10 2021 (True/False, pigeonhole) If there are 800 people in a room, then at least 3 of them are guaranteed to have the same birthday.

真题 6.11 2019 秋冬 Let $S \subseteq \{1, 2, 3, \ldots, 99, 100\}$ where $|S| = 62$. Prove that there exist $x, y \in S$ such that $x - y = 23$.

真题 6.12 2003 (True/False, pigeonhole) There are 20 students in a class. If each student has 0 or more of the other students in this class as friends, then at least 2 students have the same number of friends.

真题 6.13 2023 (True/False, pigeonhole) At least two people share the same number of hairs in Hangzhou.

8 高级计数 Advanced Counting

第 8 章 高级计数

Advanced Counting：递推关系 · 生成函数 · 容斥原理 8

知识精要 建立递推、解递推、生成函数三件事互相印证；容斥与错位排列是固定套路。

本章主线

1. **递推关系 (Recurrence Relations)**：建模、线性齐次/非齐次常系数递推的求解。
2. **生成函数 (Generating Functions)**：定义、运算、解递推与计数。
3. **容斥原理 (Inclusion–Exclusion)**：一般形式、满射计数、错位排列 (derangement)。

8.1 线性常系数递推的求解

线性齐次：特征方程法 $a_n = c_1 a_{n-1} + \cdots + c_k a_{n-k}$，特征方程 $r^k = c_1 r^{k-1} + \cdots + c_k$。
不同根 $r_1, \ldots, r_k$：$a_n = \sum \alpha_i r_i^n$；
重根 r 重数 m：贡献 $(\alpha_0 + \alpha_1 n + \cdots + \alpha_{m-1} n^{m-1}) r^n$。
常数 α_i 由初值确定。Fibonacci：$r^2 = r + 1 \Rightarrow a_n = \frac{1}{\sqrt{5}}(\varphi^n - \psi^n)$。

线性非齐次：通解 特解形式按 $F(n)$ 选取：$F(n)$ 为 n 的 t 次多项式 $\Rightarrow$ 试 t 次多项式；$F(n) = s \cdot a^n \Rightarrow$ 试 Ca^n。
若 a 恰为特征根（重数 m），特解需**乘** n^m。代回定系数。

8.2 生成函数与容斥

生成函数 (Generating Functions) $G(x) = \sum_{n \geq 0} a_n x^n$。常用：$\dfrac{1}{1-x} = \sum x^n$，$\dfrac{1}{(1-x)^k} = \sum \binom{n+k-1}{n} x^n$，$(1+x)^\alpha = \sum \binom{\alpha}{n} x^n$（广义二项式系数）。
运算：移位、求导对应 $\sum n a_n x^n = xG'(x)$；解递推时两边乘 x^n 求和得 G 的封闭式，再展开取系数。

容斥原理与错位排列 $\left|\bigcup_{i=1}^{n} A_i\right| = \sum |A_i| - \sum |A_i \cap A_j| + \cdots + (-1)^{n+1} |A_1 \cap \cdots \cap A_n|$，共 $2^n - 1$ 项。
满射计数：$|A| = m \to |B| = n$ 的满射数 $= \sum_{k=0}^{n} (-1)^k \binom{n}{k} (n-k)^m$。
错位排列 (derangement)：$D_n = n! \sum_{k=0}^{n} \dfrac{(-1)^k}{k!}$，满足 $D_n = (n-1)(D_{n-1} + D_{n-2})$。

递推与生成函数对照 判断“线性齐次常系数”三要素缺一不可：系数为**常数**（$D_n = (n-1)(\cdots)$ 系数含 n，**非**常系数）、各项次数为一次、且齐次（无额外 $F(n)$）。

✎ **真题巩固** 高级计数 · 递推建模与求解、生成函数、容斥

真题 8.1 2025 Let G_n be the graph consisting of a path with n vertices together with a new vertex adjacent to each of these n vertices, so G_n has $n+1$ vertices. Let a_n denote the number of spanning trees in G_n with $a_1 = 1$. (1) Calculate a_2. (2) Prove that $a_n = 3a_{n-1} - a_{n-2}$ for $n \geq 3$. (3) Solve the recurrence for a_n. (4) Given Fibonacci numbers f_n with $f_1 = f_2 = 1$, use induction to show that $a_n = f_{2n}$.

真题 8.2 2021 Let b_n denote the number of binary strings of length n that contain 101 as a substring, and $B(x) = \sum_{n=1}^{\infty} b_n x^n$. (a) Determine the values of b_1, b_2, b_3, b_4, b_5. (b) Derive an explicit closed-form expression for $B(x)$. *(Hint: set up recurrence relations.)*

真题 8.3 2019 Let a_n be the number of strings of length n consisting of the characters $0, 1, 2$ with no consecutive 0's. (1) Find a recurrence relation for a_n and give the necessary initial condition(s). (2) Find an explicit formula for a_n by solving the recurrence relation in part (1).

真题 8.4 2019 秋冬 (a) Let a_n denote the number of integer solutions to $5x_1+4x_2+x_3+x_4 = n$ with $0 \leq x_1,\ 0 \leq x_2,\ 0 \leq x_3 \leq 4,\ 0 \leq x_4 \leq 3$. (i) Find the generating function $G(x) = \sum_{n=0}^{\infty} a_n x^n$. (ii) Obtain a formula for a_n explicitly.
(b) Let b_n be the number of binary strings of length n that contain the substring 000. (i) Give b_4, b_5, b_6. (ii) Write a recurrence relation for b_n. (iii) Solve the recurrence relation for b_n.

真题 8.5 2020 春夏 (Fill in the blank) The particular solution of $a_n = 4a_{n-1} - 3a_{n-2} + 4^n$ is _____.

真题 8.6 2020 春夏 (Fill in the blank) If $G(x)$ is the generating function for $a_0, a_1, a_2, a_3, \ldots$, then the generating function for $0, a_1, 2a_2, 3a_3, \ldots$, in terms of $G(x)$, is _____.

真题 8.7 2020 春夏 Compute the number of ways that the digits $0, 1, 2, 3, 4, 5, 6, 7, 8, 9$ are arranged so that the first 4 digits are not in their original positions.

真题 8.8 2025 (Fill in the blank) Write the first 6 terms of the sequence determined by the generating function $\dfrac{1-2x}{1+2x}$.

真题 8.9 2023 Let a_n denote the number of ternary strings of length n containing 2 consecutive 0's (the string contains only $0, 1, 2$). (a) Find the recurrence relation for a_n and the initial conditions. (b) Solve the recurrence using characteristic roots. (c) Solve the recurrence using generating functions.

真题 8.10 2003 (Fill in the blank) Suppose $|A| = 5$ and $|B| = 3$. The number of onto functions (surjections) $f : A \to B$ is _____.

真题 8.11 2003 (Fill in the blank) The value of the extended binomial coefficient $\binom{-3}{3}$ is _____.

真题 8.12 2003 (True/False) There are $2^n - 1$ terms in the formula of the principle of inclusion–exclusion for the union of n sets, for every integer n.

真题 8.13 2025 (True/False) $\binom{0.5}{3} \times \binom{0.5}{4} > 0$.

9 关系 Relations

第 9 章 关系

Relations：性质与闭包 · 等价关系 · 偏序与 Hasse 图

9

知识精要 两条主线——等价关系对应划分，偏序对应 Hasse 图；五种性质的计数与判别是基础。

本章主线

1. **关系与表示**：0–1 矩阵、有向图、关系复合与幂。
2. **五种性质**：自反、对称、反对称、传递（及其计数）。
3. **闭包 (Closures)**：自反/对称/传递闭包、Warshall 算法。
4. **等价关系**：等价类与划分一一对应。
5. **偏序 (Partial Orders)**：Hasse 图、极大/极小/最大/最小元、上下确界、格、拓扑排序。

9.1 五种性质与闭包

关系的五种性质与计数 设 $|A| = n$。**自反 (reflexive)** $\forall a, (a,a) \in R$；**对称 (symmetric)** $(a,b) \in R \Rightarrow (b,a) \in R$；**反对称 (antisymmetric)** $(a,b),(b,a) \in R \Rightarrow a = b$；**传递 (transitive)** $(a,b),(b,c) \in R \Rightarrow (a,c) \in R$。
计数：全部关系 2^{n^2}；自反 2^{n^2-n}；对称 $2^{n(n+1)/2}$；反对称 $2^n \cdot 3^{n(n-1)/2}$；自反且对称 $2^{n(n-1)/2}$。

闭包 (Closures) 自反闭包 $R \cup \Delta$；对称闭包 $R \cup R^{-1}$；**传递闭包**（连通性矩阵）$R^* = R \cup R^2 \cup \cdots \cup R^n$，可用 **Warshall 算法** $O(n^3)$ 计算。
注意：R 传递 $\iff$ $R =$ 其传递闭包；但传递 **不**保证 $R^2 = R$。

9.2 等价关系与偏序

等价关系 $\leftrightarrow$ 划分 等价关系 = 自反 + 对称 + 传递。**等价类** $[a] = \{x \mid xRa\}$，全体等价类构成 A 的**划分 (partition)**；反之每个划分诱导一个等价关系。
n 元集上等价关系个数 = Bell 数 B_n（B_1=1, B_2=2, B_3=5, B_4=15）。

偏序、Hasse 图与界 **偏序 (poset)** = 自反 + 反对称 + 传递。**Hasse 图**省略自反与传递边、向上画。
极大/极小元：无更大/更小者；**最大/最小元**：与所有元可比且最大/最小（唯一，若存在）。子集 B 的**上界/下界**、最小上界 (lub/join) 与最大下界 (glb/meet)。**格 (lattice)**：任两元都有 lub 与 glb。
拓扑排序：将偏序扩成与之相容的全序。

等价关系 vs 偏序的“假命题” 两个等价关系的**并**一般**不是**等价关系（传递性可能被破坏）；两个传递关系的并未必传递。**唯一极大元**在**有限**偏序中是最大元，但无限偏序中未必。空集上的关系是等价关系（平凡满足）。

✎ **真题巩固** 关系 · 性质与计数、闭包、等价关系、偏序与 Hasse 图

真题 9.1 2025 (True/False) The union of two equivalence relations is also an equivalence relation.

真题 9.2 2025 (Multiple choice) For any relation T on a set, let $t(T)$ be its transitive closure and $s(T)$ its symmetric closure. Let R, S be relations on a set A. Which of the following are true? (A) $t(R \cup S) = t(R) \cup t(S)$; (B) $t(R \cap S) = t(R) \cap t(S)$; (C) $s(R \cup S) = s(R) \cup s(S)$; (D) $s(R \cap S) = s(R) \cap s(S)$.

真题 9.3 2025 For the poset $(\{2, 3, 5, 6, 12, 20, 27, 60, 108\}, |)$: (1) Draw the Hasse diagram. (2) Find the maximal elements. (3) Find the least element. (4) Find all upper bounds of $\{3, 5\}$. (5) Determine whether the underlying undirected graph of the Hasse diagram has a Hamilton path. (6) Use topological sorting to order the elements (if multiple, choose the lexicographically smallest sequence).

真题 9.4 2019 (True/False) If a relation R on a nonempty set A is transitive, then $R^2 = R$.

真题 9.5 2019 (True/False) The Hasse diagram for the partial ordering $(\{1, 2, 3, 4, 5, 6, 7, 8, 9\}, |)$ is not a tree.

真题 9.6 2019 (Fill in the blank) Suppose $A = \{1, 2, 3\}$. There are _____ relations which are reflexive and symmetric on A; there are _____ equivalence relations on A; there are _____ partial orderings on A.

真题 9.7 2019 (1) Find the smallest partial ordering on $\{1, 2, 3\}$ that contains $(1, 1), (3, 2), (1, 3)$. (2) Find the smallest equivalence relation on $\{1, 2, 3\}$ that contains $(1, 1), (3, 2), (1, 3)$.

真题 9.8 2021 (True/False) The poset $(\{1, 2, 4, 8, 12, 16, 32\}, |)$ is a lattice, where $x \mid y$ denotes x divides y.

真题 9.9 2021 (True/False) Let $(S, \preceq)$ be a partially ordered set. If there is a unique maximal element a of S, then a is the greatest element of S.

真题 9.10 2021 Let A be a set with n elements and $B = \{a, b, c\}$. (a) How many different symmetric relations on A? (b) How many different anti-symmetric relations on A? (c) How many binary relations on A are both symmetric and antisymmetric? (d) How many different equivalence relations on B? (e) How many different partial order relations on B? (f) Is there a binary relation R on B that is both an equivalence relation and a partial order? Give an example or show none exists.

真题 9.11 2020 春夏 (True/False) Let R, S be relations on a nonempty set A. If R and S are transitive, then so is $R \cup S$.

真题 9.12 2020 春夏 (Fill in the blank) Suppose $|A| = 3$. There are _____ binary relations on A; among all binary relations on A, there are _____ equivalence relations on A.

真题 9.13 2020 春夏 Let A be the set of months in the year. Month x is related to month y (xRy) if x and y begin with the same letter (e.g. January R June). (1) Show that R is an equivalence relation on A. (2) Find its equivalence classes. (3) Into how many non-empty sets is A partitioned by R?

真题 9.14 2019 秋冬 (True/False) There is a binary relation R on a set A such that R is both an equivalence relation and a partial order on A.

真题 9.15 2019 秋冬 (True/False) Suppose $(A, \preceq)$ is a finite nonempty poset. Then A has a minimal element.

真题 9.16 2019 秋冬 Define a binary relation R on $\mathbb{R} \times \mathbb{R}$ by $(x, y)R(a, b)$ if and only if $y - b = 5(x - a)$. (a) Show that R is an equivalence relation on $\mathbb{R} \times \mathbb{R}$. (b) Find the equivalence classes of $(0, 0)$ and $(1, -2)$. (c) Describe the geometric meaning of the equivalence class $[(a, b)]$.

真题 9.17 2003 (True/False) There are two different equivalence relations on a set with two elements.

真题 9.18 2003 (True/False) Let R be a relation on a set A. R equals its transitive closure if and only if R is transitive.

真题 9.19 2003 (Fill in the blank) The number of reflexive and symmetric relations on a set A with 5 elements is _____.

真题 9.20 2003 Suppose $A = \{2, 4, 6\}$. Let R be the relation on $A \times A$ where $\big((a, b), (c, d)\big) \in R$ means $a \leq c$ and $b \mid d$. (a) Show that R is a partial order relation. (b) Draw the Hasse diagram for the relation. (c) Find the maximal, minimal, greatest, least elements of the poset $(A \times A, R)$. (d) If $B = \{(2, 4), (4, 2)\}$, what are the upper bounds and lower bounds of B? Which are the least upper bound and greatest lower bound?

真题 9.21 2023 Extend $\{(1, 2), (3, 3), (4, 1)\}$ into the smallest equivalence relation (on a suitable set containing $1, 2, 3, 4$).

10 图 Graphs

第 10 章 图

Graphs：度数与同构 · 连通性 · 欧拉/哈密顿 · 平面图与着色 10

知识精要 握手定理、欧拉公式、欧拉/哈密顿判据、着色与平面图是反复考的“硬核”工具。

本章主线

1. **基本概念**：度数与握手定理、特殊图族 $K_n, C_n, W_n, Q_n, K_{m,n}$、正则图、子图。
2. **表示与同构**：邻接/关联矩阵、A^k 数路径、同构不变量。
3. **连通性**：连通分量、割点割边、强/弱连通。
4. **欧拉与哈密顿**：欧拉回路/通路判据、Dirac/Ore 充分条件。
5. **最短路、平面图、着色**：Dijkstra、欧拉公式、Kuratowski、色数与四色定理。

10.1 度数、表示与连通性

握手定理与特殊图 **握手定理**：$\sum_v \deg(v) = 2|E|$，故奇度顶点个数为偶。有向图 $\sum \deg^+ = \sum \deg^- = |E|$。
K_n：$\binom{n}{2}$ 边；C_n：环；W_n：C_n 加中心；Q_n：超立方体；$K_{m,n}$：完全二部图，mn 边。
二部图 (bipartite) $\iff$ 无奇圈 $\iff$ 可 2-着色。**邻接矩阵幂**：A^k 的 (i,j) 元为长度 k 的 $i \to j$ 通路数。

10.2 欧拉、哈密顿与最短路

欧拉与哈密顿判据 **欧拉回路 (Euler circuit)**：连通且**所有**顶点度为偶；**欧拉通路**：恰有 0 或 2 个奇度顶点。
哈密顿：无简单充要判据。充分条件：Dirac（$n \geq 3$ 且每点 $\deg \geq n/2$）、Ore（任两不相邻点 $\deg(u) + \deg(v) \geq n$）。
最短路用 **Dijkstra 算法**（非负权）。

平面图与着色 **欧拉公式**：连通平面图 $v - e + f = 2$；k 个连通分量时 $v - e + f = 1 + k$。
推论：简单连通平面图 $e \leq 3v - 6$；无三角形时 $e \leq 2v - 4$。
Kuratowski：图可平面 $\iff$ 不含 K_5 或 $K_{3,3}$ 的细分。
色数 (chromatic number) χ：$\chi(K_n) = n$，$\chi(C_n) = 2$（n 偶）或 3（n 奇），二部图 $\chi = 2$。
四色定理：平面图 $\chi \leq 4$。

图论常见“假命题” $\deg^+ = \deg^-$ 的弱连通图未必强连通；v, e, r 相同的平面图未必同构（这些只是同构**必要**不变量）；色数下界要靠**含奇圈/团**论证。证连通常用**反证 + 两分量取点**，证哈密顿常用 Ore/Dirac。

✎ **真题巩固** 图 · 度数与连通、欧拉/哈密顿、平面图、着色与综合证明

真题 10.1 2025 (True/False) Suppose G is a bipartite graph with bipartition (V_1, V_2). Then the sums of the degrees of vertices in the two partitions V_1 and V_2 are equal.

真题 10.2 2025 (Fill in the blank) For which m, n is $K_{m,n}$ planar? $(m \leq n)$

真题 10.3 2025 (Fill in the blank) The edge connectivity of a graph G, denoted $\lambda(G)$, is the minimum number of edges in an edge cut of G (a set E' of edges whose removal disconnects G). Find $\lambda(K_{m,n})$, where m, n are positive integers.

真题 10.4 2025 For a connected simple graph, define: *distance* between two distinct vertices = length of the shortest path between them; *eccentricity* of v = maximum distance from v to any other vertex; *radius* $r(G)$ = minimum eccentricity; *diameter* $d(G)$ = maximum eccentricity. For any connected simple graph G, determine whether $r(G) \leq d(G) \leq 2r(G)$ is always true, and explain why.

真题 10.5 2025 An undirected simple graph is regular if every vertex has the same degree; it is k-regular if every vertex has degree k. (1) Prove that every 2-regular graph is a disjoint union of cycles. (2) Draw two non-isomorphic 3-regular graphs, each with 6 vertices. (3) For each graph in (2), determine whether it is planar. (4) For each graph in (2), find its chromatic number.

真题 10.6 2019 (True/False) The wheel W_n is not a bipartite graph for every $n \geq 3$.

真题 10.7 2019 (True/False) A weakly connected directed graph with $\deg^+(v) = \deg^-(v)$ for all vertices v is not always strongly connected.

真题 10.8 2019 (Fill in the blank) If G is a planar connected graph with 10 vertices, each of degree 4, then G has _____ regions.

真题 10.9 2019 Let G be a planar simple graph containing no triangles; let e, v be its numbers of edges and vertices. Prove: (1) $e \leq 2v - 4$. (2) G has a vertex of degree at most 3. (3) $\chi(G) \leq 4$, where $\chi(G)$ is the chromatic number. (You cannot use the four color theorem.)

真题 10.10 2019秋冬 (True/False) There is no known algorithm to decide if a graph has a Hamilton circuit.

真题 10.11 2019秋冬 (True/False) Every connected bipartite graph contains a circuit of even length.

真题 10.12 2019秋冬 Let $G = (V, E)$ be a simple graph with n vertices, $n \geq 3$, and $d(u) + d(v) \geq n$ for every $u, v \in V$. Prove: (a) G is connected. (b) G is a Hamilton graph.

真题 10.13 2021 (True/False) All simple complete graphs with at least 3 vertices are Euler graphs.

真题 10.14 2021 Let G be a simple graph with n vertices and k connected components. (a) What is the minimum possible number of edges of G? (b) What is the maximum possible number of edges of G?

真题 10.15 2020春夏 (True/False) If $G = (V, E)$ is a simple connected non-planar undirected graph, then $|V| + |E| \geq 15$.

真题 10.16 2020春夏 (True/False) The chromatic number of a simple connected non-bipartite undirected graph is no less than 3.

真题 10.17 2020春夏 (True/False) If two planar graphs G_1 and G_2 each have v vertices, e edges, and r regions, then the two graphs are isomorphic.

真题 10.18 2020春夏 (Fill in the blank) Let A be the adjacency matrix of graph G. The number of triangles (cycles with three edges) that contain v_j is _____.

真题 10.19 2020春夏 (Fill in the blank) Let G be a planar graph with k connected components, v vertices and e edges. Then Euler's Formula for this graph is _____.

真题 10.20 2003 (True/False) Each complete bipartite graph $K_{n,n}$ has a Hamilton circuit whenever $n > 1$.

真题 10.21 2003 A regular polyhedron is a polyhedron in which all faces are congruent regular polygons, with the same configuration at each vertex (e.g. cube, dodecahedron). Every regular polyhedron is isomorphic to a planar graph. Prove that there are only 5 different regular polyhedra if the length of the edges does not matter.

真题 10.22 2003 The Computer Department has 6 committees that meet once a month. How many different meeting times must be used so that no one is scheduled for two meetings at the same time, given C_1 = {Allen,Brooks,Marg}, C_2 = {Brooks,Jones,Morton}, C_3 = {Allen,Marg,Morton}, C_4 = {Jones,Marg,Morton}, C_5 = {Allen,Brooks}, C_6 = {Brooks,Marg,Morton}? Show your answer.

注：历年卷另有若干**依赖具体图形**的题（如给定加权图求最短路、给定有向图求长度为 k 的路径数与强连通分量、给定平面图求 DFS/BFS 生成树等），因需原卷配图，此处从略，可对照原卷练习。

11 树 Trees

第 11 章 树

Trees：树与计数 · 遍历与表达式 · 哈夫曼 · 生成树

11

知识精要 树是“$n-1$ 条边的连通图”；m 叉树的计数关系、遍历与哈夫曼是高频考点。

本章主线

1. **树的概念**：等价刻画、有根树、m 叉树与满 m 叉树、非同构树计数。
2. **树的应用**：二叉搜索树、决策树、前缀码与**哈夫曼编码 (Huffman)**。
3. **遍历**：前/中/后序遍历、表达式的前缀/中缀/后缀记法。
4. **生成树**：DFS/BFS 生成树、回溯法、**最小生成树** Prim/Kruskal。

11.1 树的刻画与 m 叉树

树的等价刻画与 m 叉树关系 n 个顶点的图，下列等价：是树 $\iff$ 连通且无圈 $\iff$ 连通且恰 $n-1$ 条边 $\iff$ 无圈且恰 $n-1$ 条边 $\iff$ 任两点有唯一简单路径。
满 m 叉树 (full m-ary) 有 i 个内点：顶点数 $n=mi+1$，叶数 $l=(m-1)i+1$。
高 h 的 m 叉树至多 m^h 个叶；故 $h\geq\lceil\log_m l\rceil$（平衡时取等）。

11.2 遍历、哈夫曼与生成树

遍历与表达式记法 **前序 (preorder)**：根–左–右；**中序 (inorder)**：左–根–右；**后序 (postorder)**：左–右–根。
表达式树：前序 = 前缀 (prefix/波兰) 式，中序 = 中缀式，后序 = 后缀 (postfix/逆波兰) 式。
已知中序 + 前序（或后序）可**唯一重建**二叉树。

哈夫曼编码与生成树 **哈夫曼 (Huffman)**：反复合并**频率最小**的两棵子树，得最优前缀码；平均码长 $=\sum$(频率 $\times$ 码长)。
最小生成树 (MST)：**Prim**（从一点长出，每次加最小的跨界边）；**Kruskal**（按边权升序加入不成圈的边）。

树的计数要分清“根/标号/有序” **非同构有根树**、**非同构无根树**、**有标号树**数目各不相同（如 n 个标号顶点的树有 n^{n-2} 棵，Cayley 公式）。满 m 叉树用 $n=mi+1$ 约束；判断“至多几个叶/高度”用 m 叉树界。

✎ **真题巩固** 树 · 计数、m 叉树、遍历重建、哈夫曼与生成树

真题 11.1 2025 (True/False) There exist $m\geq 2$ and $n\geq 1$ such that there are more non-isomorphic full m-ary trees with n nodes than with $n+1$ nodes.

真题 11.2 2025 (Multiple choice) Use Huffman coding to encode these letters with the given frequencies: A:0.10, B:0.22, C:0.05, D:0.15, E:0.26, F:0.07, G:0.06, I:0.09. What is the average number of bits required to encode a symbol? (A) 2.61; (B) 2.65; (C) 2.79; (D) 2.83.

真题 11.3 2025 (Multiple choice) How many non-isomorphic unrooted spanning trees are there in the wheel W_4? (A) 3; (B) 5; (C) 6; (D) 8.

真题 11.4 2025 (Fill in the blank) A labeled tree is a tree where each vertex is assigned a label; two labeled trees are isomorphic when there is a label-preserving isomorphism. There are 3 non-isomorphic trees with 3 vertices labeled with distinct integers from $\{1, 2, 3\}$. How many non-isomorphic trees are there with four vertices labeled with distinct integers from $\{1, 2, 3, 4\}$?

真题 11.5 2019 (Fill in the blank) If T is a full 3-ary tree with 10 vertices, its minimum and maximum heights are _____.

真题 11.6 2019 (Fill in the blank) Use Huffman coding to encode these symbols with given frequencies: A:0.10, B:0.20, C:0.05, D:0.15, E:0.30, F:0.12, G:0.08. The average number of bits required to encode a symbol is _____.

真题 11.7 2019 (Fill in the blank) There are _____ non-isomorphic rooted trees with 5 vertices.

真题 11.8 2019 (Fill in the blank) A binary tree has postorder traversal DEBFCA and inorder traversal DBEACF. Its preorder traversal is _____.

真题 11.9 2019秋冬 (True/False) A binary tree with height h has at most 2^h leaves.

真题 11.10 2019秋冬 Find all non-isomorphic trees with exactly 6 vertices and four or more leaves.

真题 11.11 2021 (True/False) In a binary tree with n vertices and l leaves, $2l \leq n + 1$.

真题 11.12 2021 Suppose that T is a tree of 9 vertices with a vertex of degree 6. (a) What degree sequences can T have? (b) Draw all non-isomorphic trees of 9 vertices with a vertex of degree 6.

真题 11.13 2003 (True/False) There is an undirected tree with 2 vertices of degree 4 and 3 vertices of degree 3, with the remaining vertices being leaves; then it contains 8 leaves.

真题 11.14 2003 (Fill in the blank) If there are 8 internal nodes (not leaves) in a binary tree, how many leaves can there be at most?